



Tecnológico de Monterrey

TE3003B.501 Advanced Embedded Systems

Final delivery of Challenge

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Nuevo Leon, Mexico, November 2024

Abstract
Introduction
Objective
Justification
Product Description
Vision
Needs
Target Groups
Proposed Solution
 How It Satisfies the Target Group
Development and Teamwork Methodology Update
 Tasks/Actions Carried Out to Develop the Challenge (V-Model)
 List of Actions Carried Out for the Project
 Analysis of Each Action and Its Requirements
Task Responsibility List
 Requirement Analysis
 System Design
 Subsystem Implementation
 System Integration
 Testing and Calibration
Teamwork Methodology
Work Distribution
Context of the UML (Unified Modeling Language)
Requirements Table
Use Case Diagram
 Actors
 Use Cases
 Diagram Description
Deployment Diagram
UML Activity Diagram
Peripherals and Tools
 Sensors, Actuators, and General Hardware
 Software
System Implementation
 Complete System
 Images of the Scale Tractor
Bill of Materials
Subsystems

Pins and Peripherals Used

FreeRTOS Implementation

Communication Protocols

Testing

Unit Testing Overview

Servo Motor Test

DC Motor Test

Communication Protocols

Testing

Unit Testing Overview

Servo Motor Test

DC Motor Test

Communication Protocols

Testing Area

Conclusion

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Abstract

This project focuses on the creation of a small-scale autonomous tractor prototype in collaboration with John Deere, a global leader in agricultural automation. The prototype showcases waypoint navigation capabilities using an STM32H755 dual-core microcontroller, integrated sensors (IMU and encoders), and robust communication protocols such as SPI, CAN, and UART. The tractor is designed to simulate real-world applications, emphasizing energy efficiency, scalability, and error minimization. By leveraging advanced embedded systems and real-time processing, the project addresses the challenges of modern agriculture, including labor shortages, operational costs, and environmental sustainability. Additionally, it provides a practical framework for academic exploration and technological innovation. Testing validated the system's reliability in autonomous operations, proving its potential to transform precision farming practices.

Keywords: Autonomous Tractor, Embedded Systems, Waypoint Navigation, Precision Farming, Sustainable Agriculture, Communication Protocols, Robotics Engineering.

Introduction

The agricultural industry is evolving rapidly, driven by the need for enhanced efficiency, sustainability, and automation. John Deere, a global leader in smart farming innovations, is pioneering the development of autonomous systems to support these objectives. In collaboration with John Deere, this project focuses on creating a scale model tractor that demonstrates advanced waypoint navigation. The prototype integrates new ways for embedded systems, robust communication protocols, and precise sensor technology to simulate real-world applications in modern agriculture. By addressing the growing demands for automation, this initiative underscores the transformative potential of smart farming practices.

Objective

The project aims to design and develop a small-scale tractor capable of autonomous waypoint navigation, demonstrating the application of advanced embedded systems. The specific objectives include:

1. **Seamless Waypoint Navigation:** Ensure that the tractor follows predefined paths with minimal deviation, IMU, and encoder data for accurate positioning.
2. **Integration of Communication Protocols:** Leverage SPI, CAN, UART, and I2C protocols for reliable communication between subsystems.
3. **Efficient Task Management:** Implement dual-core processing for better tasks performing and optimize resource usage.
4. **Error Minimization:** Incorporate real-time feedback mechanisms to reduce navigation errors.

By achieving these goals, the project contributes to the broader vision of John Deere and other agricultural pioneers to revolutionize farming through scalable automation.

Justification

The automation of agricultural equipment is critical to addressing labor shortages, reducing operational costs, and improving productivity. Autonomous tractors that navigate using IMU data exemplify the future of precision farming. This project is a response to these demands, demonstrating how embedded systems can seamlessly integrate hardware and software to achieve reliable, automated navigation. Collaborating with John Deere provides an opportunity to align the prototype with industry standards and real-world requirements.

Additionally, this scale model showcases:

1. **Cost-Efficient Prototyping:** Using affordable and modular components to test complex systems.
2. **Environmental Benefits:** Precise navigation reduces fuel consumption and minimizes environmental impact.
3. **Educational Value:** Offering insights into embedded systems for students and professionals.

By meeting these needs, the project not only supports John Deere's goals but also highlights the value of innovation in agriculture and common learning in between Robotics Engineering.

Product Description

The proposed system is a scale tractor prototype equipped with:

1. **Dual-Core Microcontroller:** The STM32H755 manages multitasking operations efficiently.
2. **Integrated Sensors:** Includes, an IMU for orientation, and encoders for precise motion control.
3. **Actuators:** A servo motor for steering and a DC motor for traction, controlled via PWM signals.
4. **Communication Modules:** Wireless NRF24 modules, CAN transceivers, and SPI-based devices ensure seamless data transfer.

This tractor focuses on mimicking a full-scale autonomous vehicle, showcasing the integration of hardware and software into a functional system. The collaboration with John Deere emphasizes its practical relevance, as it aligns with their advancements in precision agriculture

Vision

The vision of this project is to bridge the gap between technological innovation and practical agricultural solutions, empowering farmers with tools that increase efficiency while promoting sustainability. The collaboration with John Deere underscores a shared commitment to smarter, more precise farming practices. The prototype serves as a proof of concept, inspiring future development of scalable technologies that meet global agricultural demands while reducing environmental footprints.

Needs

The scale tractor prototype addresses several key agricultural needs:

1. **Autonomous Navigation:** Eliminating the need for human intervention by enabling vehicles to independently traverse predesignated paths.
2. **Operational Precision:** IMU data to minimize deviations, ensuring accurate task execution.
3. **Efficient Communication:** Utilizing multiple communication protocols for stable data exchange between subsystems.
4. **Energy Efficiency:** Optimizing power usage to simulate the constraints of real-world farming environments.
5. **Adaptability:** Configurable to different routes and tasks, making it a versatile solution for diverse agricultural applications.

By addressing these needs, the prototype aligns with John Deere's vision of integrating advanced technologies into farming equipment.

Target Groups

The potential beneficiaries of this project include:

1. **Farmers and Agricultural Manufacturers:** Offering automation solutions that reduce labor and improve efficiency.
2. **Institutions and Universities:** Providing a practical model for exploring advanced embedded systems and smart farming technologies.
3. **Agricultural Technology Innovators (e.g., John Deere):** Supporting the development of scalable and commercially viable autonomous systems.

By targeting these groups, the project underscores its relevance to both academic and industrial stakeholders, fostering innovation in the agricultural sector.

Proposed Solution

The solution is a fully functional prototype of a scale tractor that autonomously navigates through waypoints. Key components include:

1. **Real-Time Navigation System:** Integrating IMU, and encoders for precise positional tracking.
2. **Multitasking Architecture:** Using dual-core STM32H755 microcontroller to prioritize tasks efficiently.
3. **Communication Framework:** Employing protocols like CAN for motor feedback and SPI for wireless modules to ensure consistent data exchange.
4. **Robust Construction:** Mechanical and electronic integration that guarantees durability and accuracy in testing conditions.

How It Satisfies the Target Group

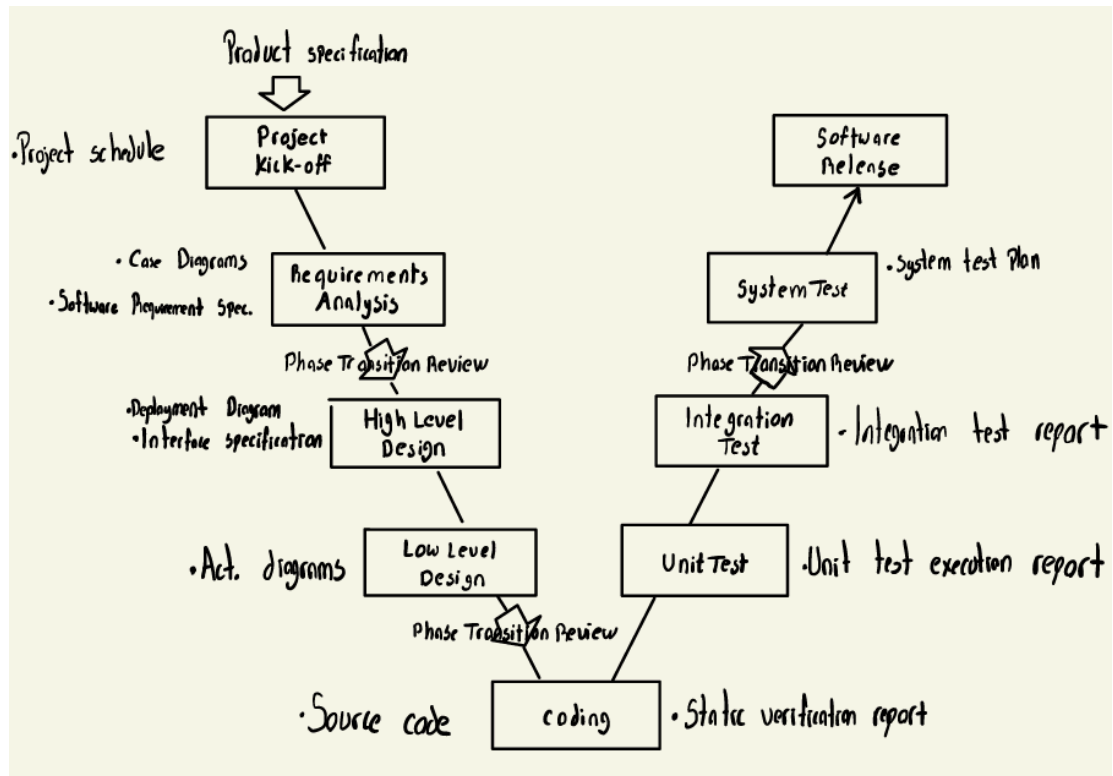
- **Farmers:** Demonstrates the reduction of labor and operational errors through autonomous navigation.
- **Researchers:** Offers a modular, scalable model for exploring communication protocols and sensor integration.
- **John Deere:** Provides a proof-of-concept system that aligns with their ongoing efforts to revolutionize agriculture through technology.

The tractor serves as a foundational step toward realizing full-scale autonomous farming machinery, addressing key challenges in precision agriculture

Development and Teamwork Methodology Update

Tasks/Actions Carried Out to Develop the Challenge (V-Model)

The development process followed the V-model, emphasizing rigorous validation and verification at each stage. Below is a comprehensive list of tasks and actions carried out, categorized by development phases:



List of Actions Carried Out for the Project

The following tasks were executed over the project timeline:

1. **Requirement Analysis and Documentation:**
 - Defined system requirements based on project goals.
 - Created a context diagram to illustrate system boundaries and external interactions.
2. **System Design:**
 - Developed a deployment diagram to represent the hardware and software architecture.
 - Created UML activity diagrams to outline workflows and state transitions.
3. **Subsystem Implementation:**
 - Assembled the hardware components, including the scale tractor's frame, motors, and sensors.
 - Configured the STM32 microcontroller and implemented the interface with UART for debugging.
 - Established communication with the IMU sensor using I2C for orientation tracking.
 - Controlled the DC motor and servo motor using PWM signals for accurate movement.
 - Integrated NRF24 wireless modules for waypoint data exchange.
4. **System Integration:**

- Combined all subsystems, ensuring seamless communication between components via CAN and SPI protocols.
- 5. **Testing and Calibration:**
 - Conducted unit testing for individual modules, such as the IMU, motors, and UART.
 - Performed integration testing to validate subsystem interactions.
 - Executed system-level testing to assess the tractor's performance in navigating waypoints.

Analysis of Each Action and Its Requirements

1. **Requirement Analysis:**
 - **Objective:** To ensure that all team members had a clear understanding of the project scope.
 - **Requirement:** Familiarity with project goals, system constraints, and John Deere's expectations.
2. **System Design:**
 - **Objective:** To provide a blueprint for hardware and software integration.
 - **Requirement:** Knowledge of UML diagramming and deployment strategies for embedded systems.
3. **Subsystem Implementation:**
 - **Objective:** To develop individual modules in isolation before integration.
 - **Requirements:**
 - Proficiency in configuring STM32 peripherals and communication protocols (UART, I2C, SPI).
 - Mechanical assembly skills to ensure stable hardware construction.
 - Expertise in controlling motors using PWM for precise navigation.
4. **System Integration:**
 - **Objective:** To ensure all subsystems work cohesively.
 - **Requirements:** Understanding of data flow between components and the ability to debug integration issues.
5. **Testing and Calibration:**
 - **Objective:** To validate functionality and optimize system performance.
 - **Requirements:**
 - Ability to design and execute unit tests for individual components.
 - Skills in troubleshooting and fine-tuning subsystems during integration.

Task Responsibility List

Requirement Analysis

- **Context Diagram Preparation:** Rodrigo led the creation of the UML context diagram to define system scope and interactions.

- **Requirements Table Development:** **Rosendo** compiled and documented functional and non-functional requirements for the project.

System Design

- **Deployment Diagram:** **Jordan** developed the deployment diagram, mapping the interactions between hardware and software components.
- **Activity Diagrams:** **Jesus** created UML activity diagrams, illustrating system workflows and state transitions.

Subsystem Implementation

- **Tractor Assembly:** **Rosendo** oversaw the mechanical assembly of the tractor, ensuring stable hardware integration.
- **UART Interface:** **Jordan** configured UART for debugging and data transmission.
- **IMU Interface:** **Jesus** integrated the IMU sensor using I2C, enabling accurate orientation tracking.
- **Motor Control:** **Rodrigo** implemented the PWM control for both the DC and servo motors.
- **Wireless Communication:** **Jordan** established communication using NRF24 modules for waypoint data exchange.

System Integration

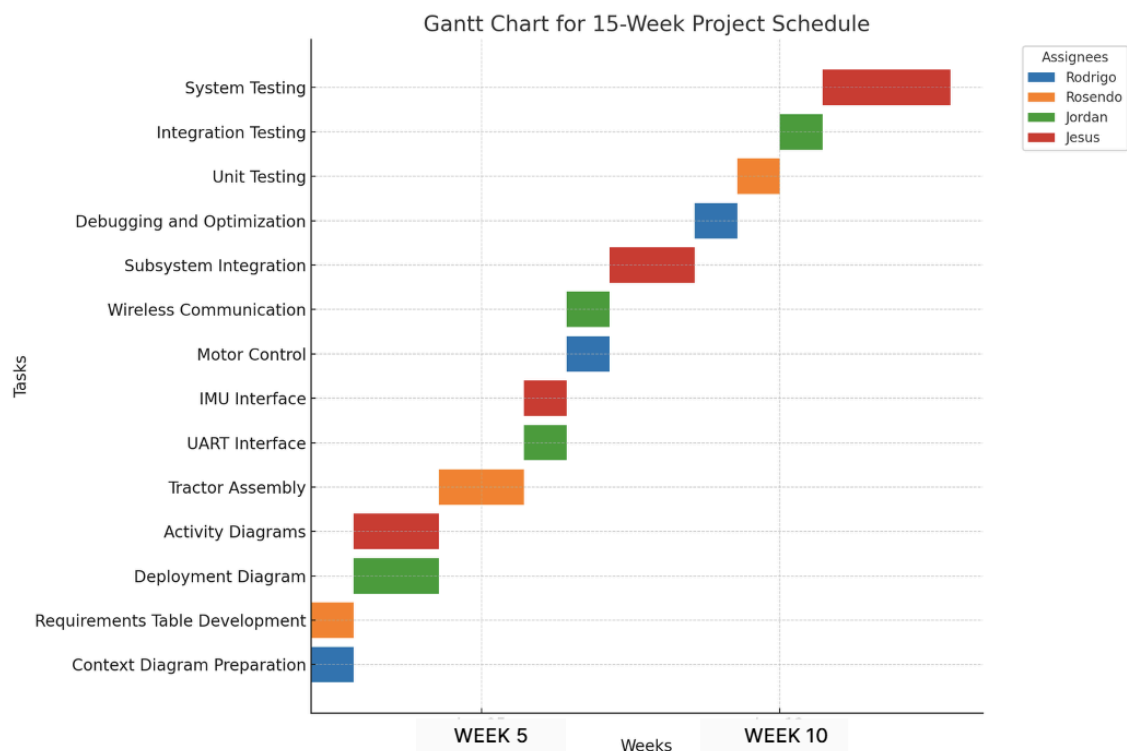
- **Subsystem Integration:** **Jesus** combined all subsystems, ensuring seamless communication between modules via CAN and SPI protocols.
- **Debugging and Optimization:** **Rodrigo** led debugging efforts during system integration, addressing latency and data flow issues.

Testing and Calibration

- **Unit Testing:** **Rosendo** conducted component-level testing to validate individual subsystems, such as the IMU and motors.
- **Integration Testing:** **Jordan** assessed interactions between subsystems, including sensor data fusion and motor control.
- **System Testing:** **Jesus** evaluated overall system performance, testing waypoint navigation in a controlled environment.

GANTT Diagram Distribution

Gantt diagram shows a better distribution of work throughout a period of 15 weeks



Teamwork Methodology

Our approach to teamwork was grounded in collaboration and efficient task allocation. At the start of the project, we held initial meetings to discuss the tasks at hand and to ensure that each team member understood the project's objectives. We adopted a method that allowed us to divide responsibilities based on each individual's strengths and expertise, which helped streamline our workflow. The complexity of the project required us to balance both individual and collaborative efforts, with constant communication to ensure synchronization across different subsystems. By adopting a clear structure and maintaining regular check-ins, we were able to stay on track despite the challenges encountered throughout the development process.

Work Distribution

The work distribution was organized according to the specific skills and availability of each team member. Each member was given clear ownership of critical tasks, while also allowing for overlap and support where needed. The key aspects of the project were divided as follows:

- Rodrigo was primarily responsible for motor control, CAN communication, and debugging tasks. His background in electrical engineering made him a perfect fit for these technical roles.

- Rosendo focused on mechanical assembly, integrating components, and conducting unit testing for hardware subsystems. His experience with hardware integration ensured that the assembly process went smoothly.
- Jordan handled the software development aspects, including UART and wireless communication protocols. His work ensured that the system's communication components were seamlessly integrated.
- Jesus led the integration of subsystems, IMU interface, and system testing. He was responsible for ensuring that all components worked together effectively and debugging any issues that arose during integration.

By aligning tasks to individual strengths, we were able to optimize productivity and ensure that the project was progressing in parallel across different phases.

Improvements on Teamwork Methodology

Reflecting on the teamwork methodology, we identified several areas where we could improve to ensure even more effective collaboration:

Buffer Time for Contingencies: In hindsight, we realized that unexpected delays—such as hardware issues or personal commitments—impacted our schedule. Moving forward, we would incorporate buffer time into the project timeline to account for unforeseen circumstances, ensuring that we have enough flexibility to adjust without compromising deadlines.

Cross-Training Team Members: While tasks were assigned based on expertise, there were instances when one member's delay in their task impacted the next steps. To mitigate this in the future, we suggest cross-training team members on tasks outside their core responsibilities. This would allow for greater flexibility and faster resolution of bottlenecks if a team member is unavailable or if another subsystem requires more attention.

Increased Use of Collaborative Tools: Although we maintained regular communication, we could improve the efficiency of our collaboration by using more advanced project management tools. Tools like Trello or Asana could have helped track progress, set clear milestones, and provide a better overview of the project's development stages, reducing reliance on meetings and emails.

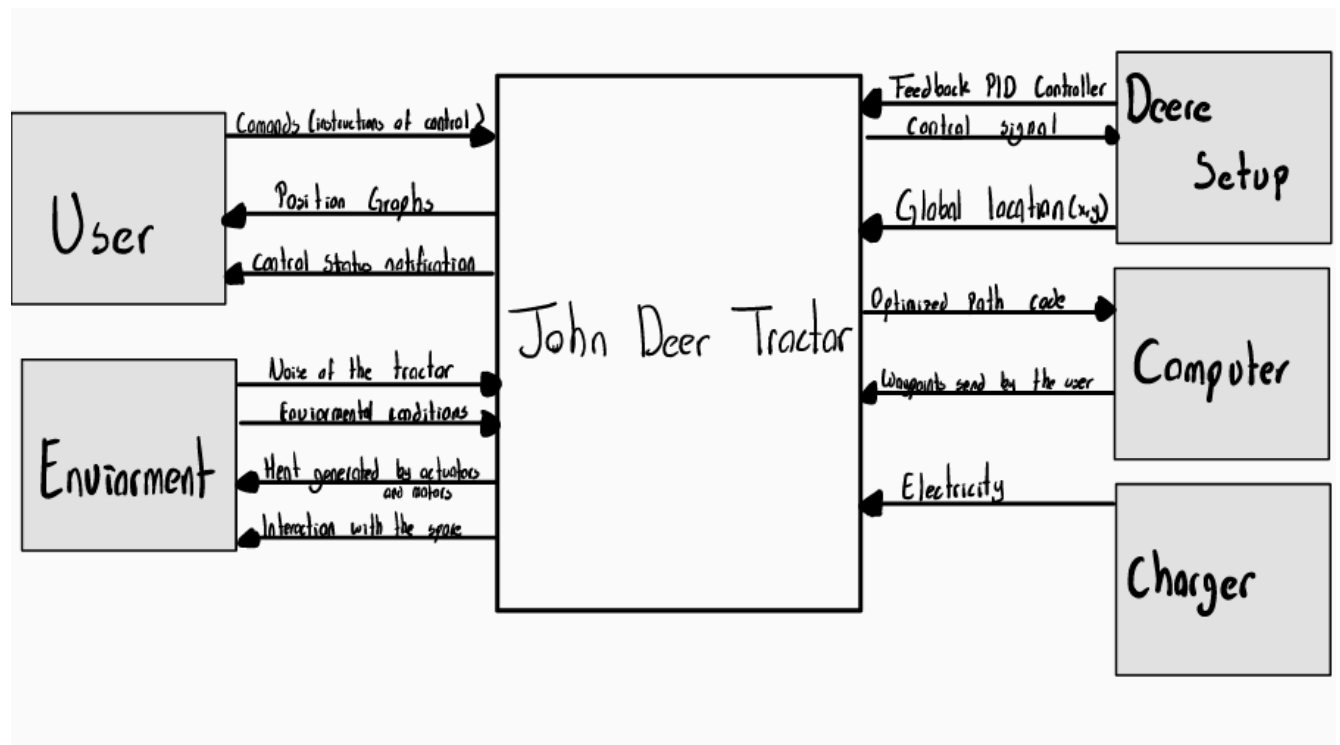
Better Coordination in Testing Phases: The testing phase often saw overlap between unit testing and system testing, causing some confusion about task prioritization. In the future, we could establish clearer distinctions between different testing stages and ensure that members are aligned on which subsystem should be tested first to avoid redundant efforts.

More Defined Task Ownership: While the tasks were well-distributed, in some cases, there were moments when responsibilities were unclear, especially during the system integration phase. Defining a clearer hierarchy of task ownership, particularly in the later stages of the project, would ensure that each member knows exactly when and how to step in for support.

These improvements would enhance our team's collaboration and ensure that we are better prepared to handle the complexity of future projects. Overall, our experience in this project taught us the importance of adaptability and communication in achieving a successful outcome.

Context of the UML (Unified Modeling Language)

UML is a standardized modeling language used to visualize, specify, construct, and document the artifacts of a system. It provides a comprehensive approach to model various aspects of a system, from architecture to behavior, allowing developers and stakeholders to have a unified view of how the system works and interacts. In the context of your project, UML diagrams, such as context diagrams and use case diagrams, are used to define the system's interactions with users and external systems. These diagrams help clarify requirements, user interactions, and how the system will behave under different conditions.



Requirements Table

The Requirements Table outlines the functional and non-functional specifications that the system must satisfy. It helps both the development team and the stakeholders understand the system's capabilities and constraints.

ID	Requirements	Type	Description
R1	Autonomous Navigation	Functional	The system must guide the tractor through a series of predefined waypoints autonomously.
R2	Sensor Integration	Functional	The system should for position tracking and IMU for orientation monitoring.
R3	Motor Control	Functional	The microcontroller must regulate the speed and direction of the DC motor and control the servo for steering.
R4	Wireless Data Exchange	Functional	The tractor must communicate with external systems using a wireless module (e.g., NRF24).
R5	Task Prioritization	Functional	The system must prioritize tasks (e.g., motor control, data processing) using an RTOS to avoid delays.
R6	Feedback to User	Functional	The system should provide real-time feedback, including position, errors, and task status, to the user.
R7	Power Efficiency	Non-Functional	The hardware and software must be optimized for low power consumption to extend operating time.
R8	Scalability	Non-Functional	The system design must allow for easy upgrades, including adding new sensors or extending communication capabilities.
R9	Robustness	Non-Functional	The tractor must withstand common field conditions such as vibrations and minor impacts.
R10	System Diagnostics	Non-Functional	The system must perform health checks on components (e.g., sensors, motors) and report malfunctions.

Use Case Diagram

Actors

1. **User (Operator/Farmer):**
 - Responsible for setting up the system, entering waypoints, monitoring operations, and responding to feedback from the tractor.
2. **External System:**
 - Includes devices or systems that provide data (e.g., waypoint coordinates) or receive updates from the tractor.

Use Cases

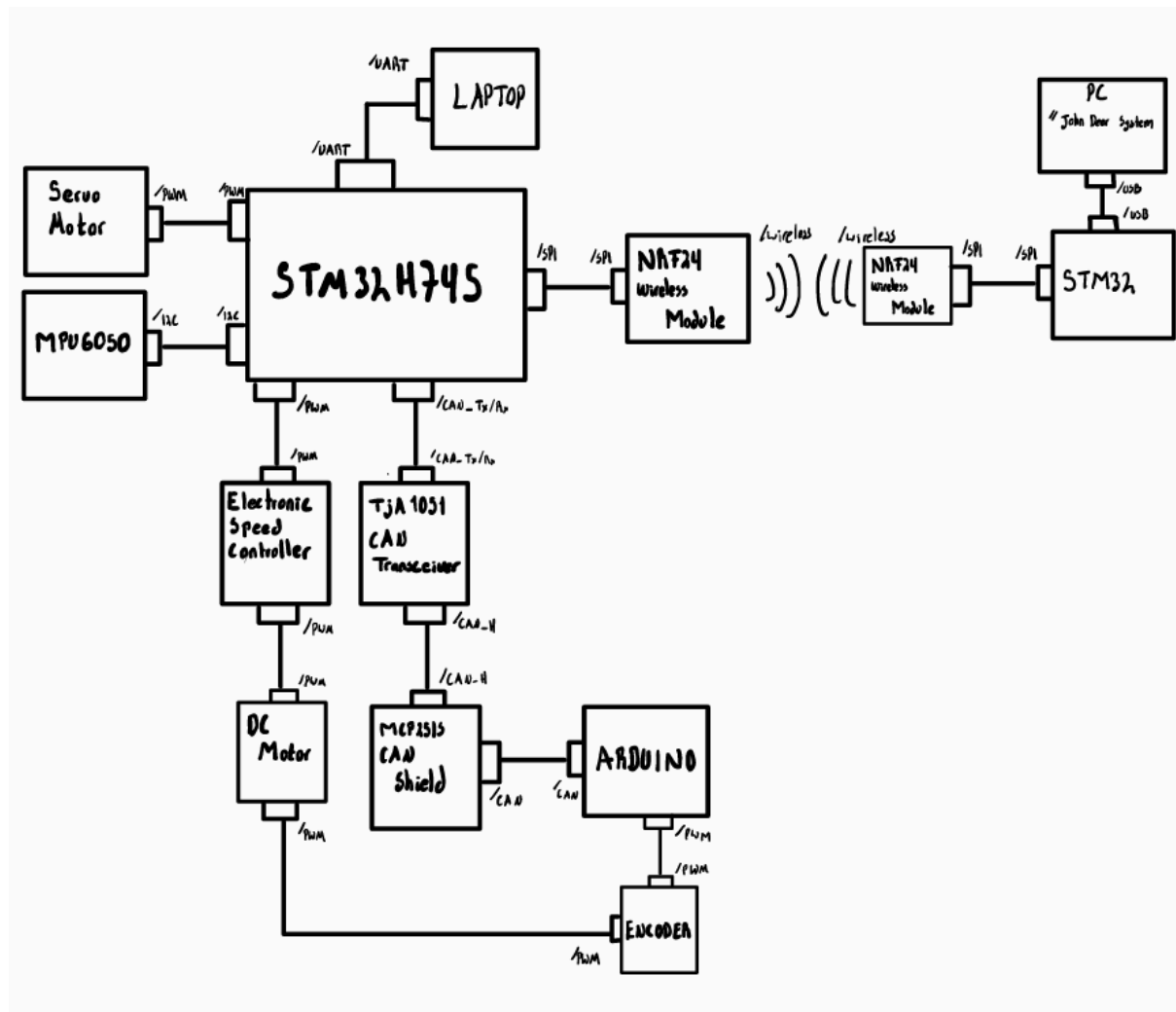
1. **Set Waypoints:**
 - The user defines the route by entering waypoints into the system.
2. **Start Navigation:**
 - The user initiates the tractor's autonomous operation after confirming the setup.
3. **Monitor Progress:**
 - The user observes the real-time position and status of the tractor during operation.
4. **Receive Error Notifications:**
 - The system alerts the user to any issues, such as path deviation, low power, or sensor failure.
5. **Control Override:**
 - The user can manually take control of the tractor if necessary (e.g., in case of emergency).
6. **System Diagnostics:**
 - The system runs self-diagnostic checks and reports component status to the user.

Diagram Description

- **Actors:**
 - **User** interacts with all six use cases (e.g., waypoint setup, monitoring, and manual override).
 - **External System** is involved in providing waypoint data and receiving status updates.
- **Interactions:**
 - The user initiates waypoint setup, receives feedback from the system, and can override navigation as needed.
 - The external system supports the tractor with input/output data to ensure smooth operation.

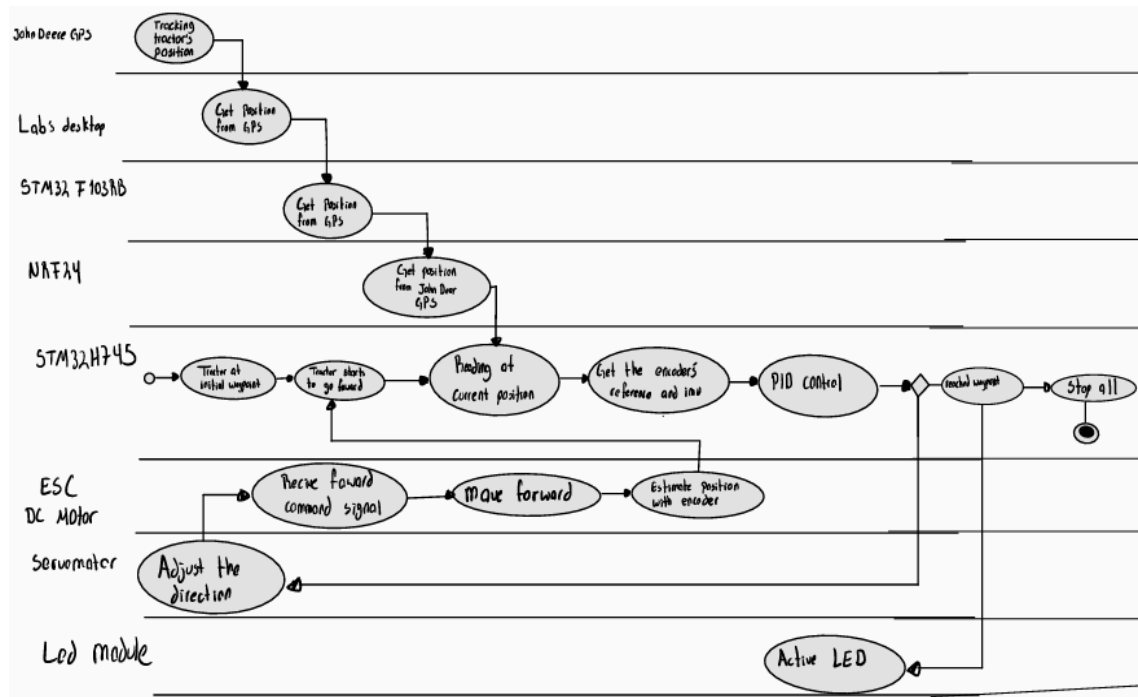
Deployment Diagram

A Deployment Diagram highlighting how components interact with each other in a system's environment. This diagram is used to visualize the architecture of the system, showing the physical configuration of devices (hardware) and how the software components are distributed across them.



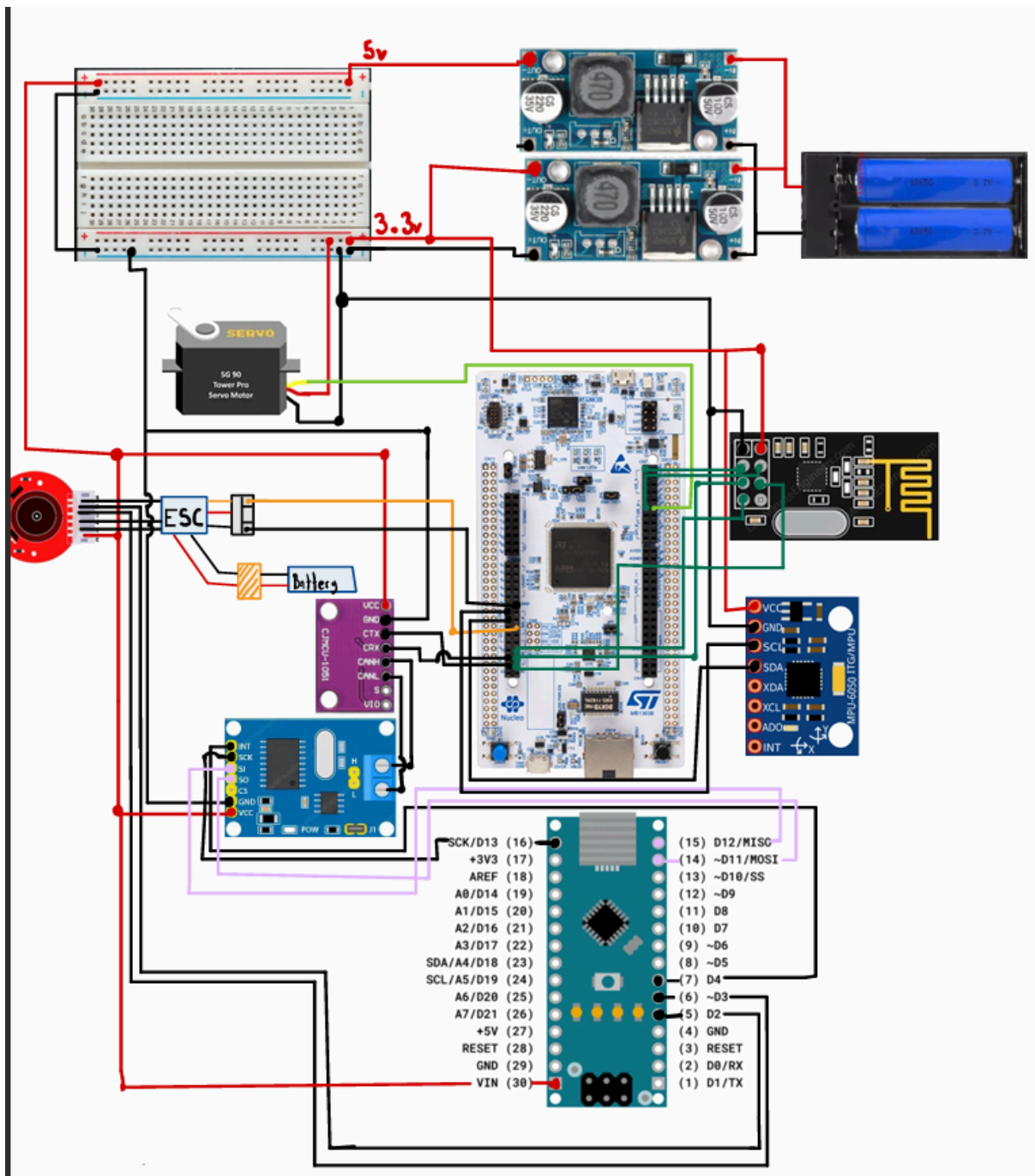
UML Activity Diagram

The behavioral diagram that models the dynamic aspects of a system. It represents workflows of actions and decisions, showing how a system processes data or carries out a task step by step. Activity diagrams are particularly useful for illustrating the flow of control and data in a system, and they can show the sequence of operations in a process or workflow.



Schematic Diagram

Shows the circuitry and layout of various electronic components, such as sensors, actuators, microcontrollers, and other components, with symbols that represent their functions. In the case of your autonomous tractor, the Schematic Diagram would provide a detailed view of how the different hardware components are connected electrically.



Peripherals and Tools

Microcontrollers

STM32H755 Microcontroller:

Justification:

- Chosen for its dual-core architecture (Cortex-M7 and Cortex-M4), enabling efficient multitasking.
- Supports advanced peripherals such as PWM, SPI, CAN, and UART, critical for sensor integration and motor control.
- Its compatibility with FreeRTOS allows real-time task management for the tractor's navigation and control system.

Arduino Nano:

Justification:

- Used as a supplementary controller to process encoder data and communicate with the main microcontroller via CAN.
- Its compact size and compatibility with the MCP2515 CAN shield made it a cost-effective option for handling specific tasks.

Sensors, Actuators, and General Hardware

Sensors and actuators enable interaction with the environment and provide the necessary control for movement and navigation. Below is the list of key components:

Components	Role	Justification
IMU (MPU6050)	Tracks orientation and motion (accelerometer and gyroscope).	Compact design, dual accelerometer-gyroscope capability, and I2C interface for efficient communication with the STM32.
DC Motor with Encoder	Drives the tractor forward or backward; encoder provides speed and position feedback.	Encoders ensure precise control of speed and position, crucial for smooth operation and accurate waypoint tracking.

Servo Motor	Controls steering of the tractor for directional adjustments.	Provides precise angular control through PWM signals, enabling real-time steering adjustments during navigation.
NRF24 Wireless Module	Facilitates wireless communication for remote waypoint updates and system monitoring.	Known for low latency and reliability, its SPI compatibility ensures stable integration with the STM32 microcontroller.
Power Supply (Batteries)	Powers the entire system, ensuring stable operation under load.	Multiple batteries provide sufficient power for high-demand components (e.g., motors), with voltage regulators ensuring appropriate supply to sensitive electronics.
Motor Driver/ESC	Controls the DC motor's speed and direction based on PWM signals from the microcontroller.	Simplifies motor control, allowing the STM32 to focus on higher-level operations while ensuring smooth motor performance under various load conditions.

Software

List and Justification

The software architecture integrates development tools, libraries, and frameworks that collectively manage the system's operation and provide debugging and testing capabilities. Here's a detailed overview:

STM32Cube IDE:

- Purpose:**
 Provides an integrated development environment for programming and debugging the STM32 microcontroller.
- Reason for Selection:**
 The IDE includes built-in support for the STM32 series, along with libraries for configuring peripherals (e.g., PWM, SPI, UART). Its debugging tools enable quick identification and resolution of software issues, making it indispensable for this project.

FreeRTOS:

- **Purpose:**
Manages task scheduling and prioritization to ensure real-time responsiveness.
- **Reason for Selection:**
FreeRTOS enables efficient multitasking by dividing the processor's time between critical tasks (e.g., motor control) and non-critical tasks (e.g., wireless communication). This ensures that the tractor remains responsive even under high workloads.

Arduino IDE:

- **Purpose:**
Used for programming the Arduino Nano, particularly for its role in CAN communication and encoder data processing.
- **Reason for Selection:**
Known for its simplicity and extensive library support, the Arduino IDE allows quick development and testing of firmware for the Nano.

System Implementation

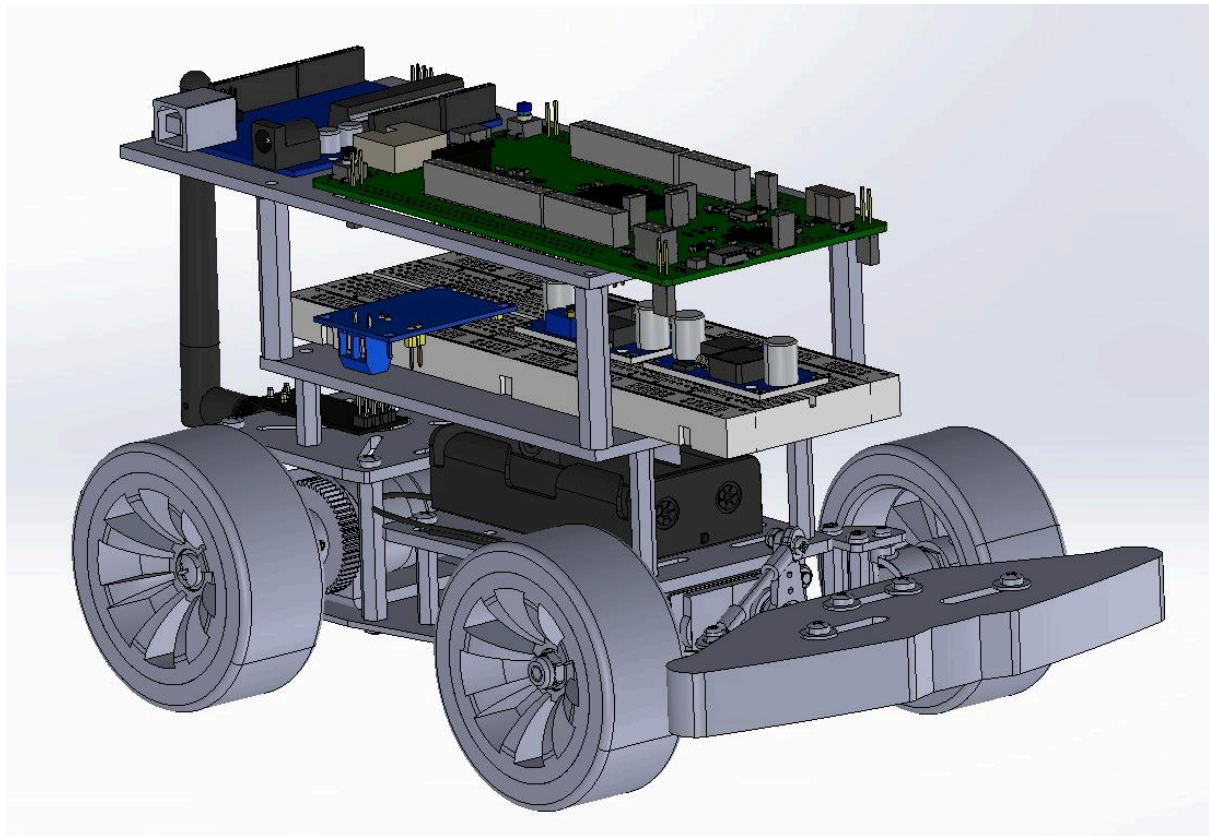
PreAssembled Tractor

The pre-assembled tractor consists of a robust and compact design that integrates key components such as the STM32 microcontroller, Communication modules, DC motor with encoder, and servo motor for steering. The chassis is engineered to securely house all components while maintaining easy access for testing and adjustments. The wiring and connections were carefully planned to avoid interference, ensuring smooth operation. This phase validated the mechanical and electronic assembly, ensuring the tractor was ready for testing.

The following components were included in the pre-assembled stage:

- **Base Chassis:** Supports all hardware components.
- **DC Motor and Servo Motor:** Installed for propulsion and steering control.
- **Sensors:** Encoder was mounted and connected to the microcontroller.
- **Power System:** Batteries and voltage regulators were configured to supply stable power to the tractor.

Here is a design of the pre assembled tractor for a better compact design that integrates key components:



Complete System

The system integrates hardware and software components to create an autonomous scale tractor capable of navigating predefined waypoints.

1. Hardware Integration:

- The STM32H755 microcontroller serves as the central control unit, managing communication with sensors, actuators, and external systems.
- Sensors like the IMU provide continuous feedback on position and orientation, while motors and servos adjust movement based on calculated trajectories.
- Wireless communication modules ensure the system remains connected to external devices for updates and remote monitoring.

2. Software Architecture:

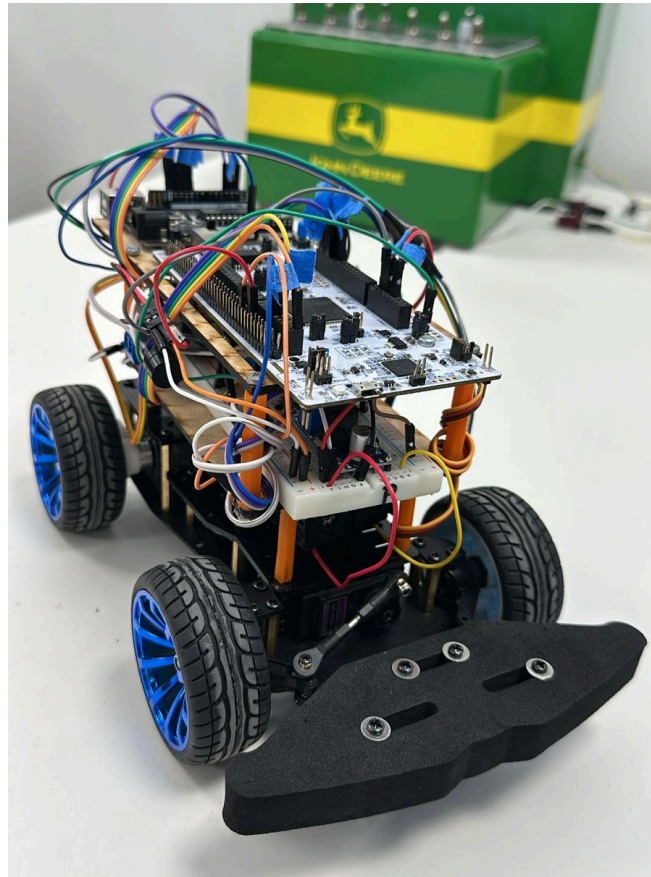
- The software is built around FreeRTOS, which allows tasks such as motor control, sensor data fusion, and communication to run concurrently without interfering with each other.
- Firmware algorithms process sensor data in real time, generating precise motor control commands to navigate the tractor through waypoints.

3. Communication Protocols:

- SPI is used for wireless communication, UART, and I2C for IMU integration. The CAN protocol connects auxiliary modules like the Arduino Nano for enhanced functionality.

Images of the Scale Tractor

- **Fully Assembled Tractor:** Highlighting the placement of sensors, motors, and the microcontroller.



Bill of Materials

The project required carefully selected components to balance performance and cost. Below are the major materials used:

1. **Microcontroller:** STM32H755 for its dual-core functionality and support for multiple peripherals.
2. **Motors:** A DC motor with an encoder for propulsion and a servo motor for steering.
3. **GPS Module:** Provides real-time location data.
4. **Wireless Communication:** NRF24 for efficient and reliable data exchange.
5. **Power Supply:** Batteries and voltage regulators to meet the system's diverse voltage requirements.
6. **Miscellaneous:** Connectors, wires, and mounting hardware for assembly.

The materials were chosen based on their compatibility, reliability, and ability to meet the project's performance requirements.

Bill of Materials (BOM) Table

Components	Quantity	Description	Estimated Cost (MXN)
STM32H755 Microcontroller	1	Dual-core microcontroller for real-time task management.	\$820
DC Motor with Encoder	1	Propulsion motor with feedback for precise control.	\$180
Servo Motor	1	Controls steering angle.	\$180
Arduino UNO	1	Secondary microcontroller for auxiliary tasks and sensor integration.	\$250
TJA1051 CAN Transceiver	1	Interfaces CAN communication between devices.	\$305
NRF24 Wireless Module	1	Handles wireless data exchange.	\$179
Power Supply (Batteries)	1 set	Provides stable power to all components	\$100
Motor Driver/ESC	1	Controls DC motor speed and direction.	\$300
Miscellaneous (Wires, PCB, 3D printed chassis)	1	Connectors, wires, and additional mounting hardware.	\$600
Rechargeable Battery	1	Powers all hardware components of the system.	\$200
RC Vehicle Kit	1	Provides the chassis and base for mounting components.	\$5000
Voltage Regulator (LM2596)	2	Regulates voltage for components with varying power requirements.	\$60
Total			\$8174

Subsystems

The tractor's functionality is divided into several interconnected subsystems, each responsible for specific tasks:

1. **Navigation Subsystem:**
Utilizes encoder data to determine the tractor's position and adjust its movement along predefined waypoints.
2. **Control Subsystem:**
Manages propulsion and steering using PWM signals generated by the STM32 microcontroller to control the DC motor and servo motor.
3. **Communication Subsystem:**
Ensures data exchange between internal components (e.g., sensors, motors) and external systems using protocols like SPI, UART, and CAN.
4. **Power Subsystem:**
Distributes stable voltage to all components, ensuring continuous operation under varying loads.

Pins and Peripherals Used

The STM32 microcontroller interfaces with several components, each requiring specific pins for communication and control:

- **PWM:** For controlling the DC motor (propulsion) and servo motor (steering).
- **UART:** For communication with the GPS module.
- **SPI:** For wireless communication via the NRF24 module.
- **CAN:** For connecting auxiliary controllers like the Arduino Nano.

The pin assignments were carefully planned to optimize the use of the STM32's peripherals and maintain reliable connections.

2. **SPI:**
Interfaces with the NRF24 wireless module, enabling reliable and efficient data transmission.
3. **CAN:**
Integrates auxiliary modules like the Arduino Nano, ensuring stable communication over a robust bus system.

These protocols were chosen for their reliability, speed, and compatibility with the project's hardware.

Testing

Unit Testing Overview

Unit testing focused on verifying the functionality of individual components before integrating them into the system. The primary goal was to identify and resolve issues early, ensuring seamless operation during system-level testing.

Servo Motor Test

- **Objective:** Validate the servo motor's ability to adjust steering angles in response to PWM signals.
- **Procedure:** Input PWM signals of varying duty cycles and observe the motor's movement.
- **Outcome:** The servo responded accurately to changes in PWM, providing smooth and precise steering adjustments.

DC Motor Test

- **Objective:** Test the DC motor's speed control and encoder feedback.
- **Procedure:** Apply varying PWM signals to the motor and monitor the encoder output.
- **Outcome:** The motor responded to speed changes, and encoder feedback matched the expected results.

Communication Protocols

- **SPI and Wireless Communication:** Data packets were sent and received reliably via the NRF24 module.
- **CAN Communication:** Stable data transmission between the STM32 and Arduino Nano was verified.
- **UART Communication:** GPS data was received accurately, providing continuous positional updates.

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- **Procedure:** Apply varying PWM signals to the motor and monitor the encoder output.
- **Outcome:** The motor responded to speed changes, and encoder feedback matched the expected results.

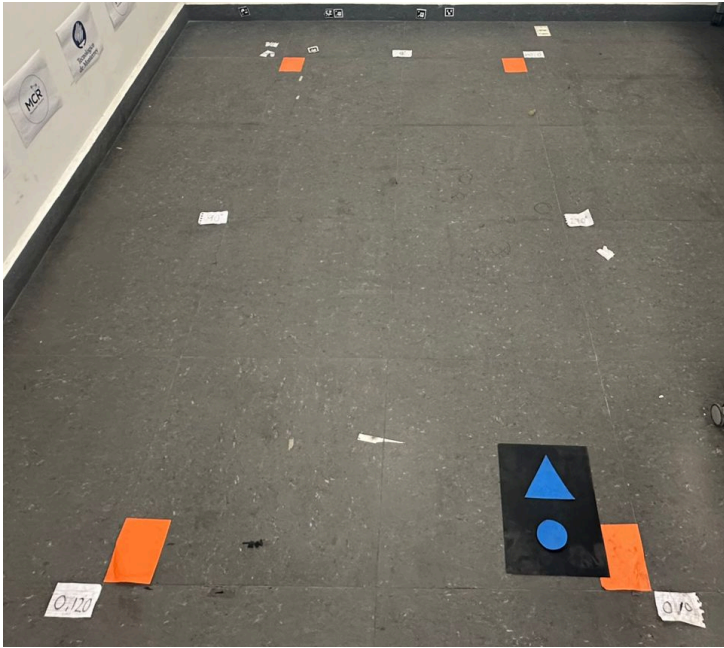
Communication Protocols

- **SPI and Wireless Communication:** Data packets were sent and received reliably via the NRF24 module.
- **CAN Communication:** Stable data transmission between the STM32 and Arduino Nano was verified.
- **UART Communication:** GPS data was received accurately, providing continuous positional updates.

Test ID	Component	Test Description	Expected Result	Status
UT-01	GPS Module	Verify that the GPS provides accurate positional data in real-time.	Latitude and longitude data are displayed correctly.	✓ Passed
UT-02	DC Motor with Encoder	Confirm that the encoder provides speed and position feedback while the motor runs.	Encoder data matches motor movement precisely.	✓ Passed
UT-03	Servo Motor	Validate the servo motor responds to PWM signals and adjusts the steering angle.	Steering angle changes as per PWM input.	✓ Passed
UT-04	NRF24 Wireless Module	Check if the wireless module can send and receive data packets successfully.	Data is transmitted and received without errors.	✓ Passed
UT-05	STM32 Microcontroller	Ensure that the STM32 processes data and generates PWM signals for motor control.	PWM signals are generated based on input data.	✓ Passed
UT-06	Power Supply	Test the power supply to verify stable voltage and current output under load.	Voltage remains stable within acceptable limits.	✓ Passed

Testing Area

The system was tested in a controlled environment simulating real-world conditions. The testing area included obstacles and varying terrain to challenge the navigation algorithms and assess the tractor's ability to adapt to dynamic environments. The tests demonstrated the system's reliability and precision in maintaining its path and adjusting to external factors.



Conclusion

The project successfully achieved its objectives, developing a functional prototype of an autonomous tractor capable of navigating predefined waypoints with high accuracy and reliability. By integrating a dual-core STM32H755 microcontroller, IMU sensors, and advanced communication protocols, the system demonstrated the potential of embedded systems in addressing real-world agricultural challenges. The tractor's performance validated its robustness and adaptability to varying conditions, aligning with industry standards and sustainable farming goals.

This initiative also highlighted the importance of teamwork and interdisciplinary collaboration in achieving complex engineering goals. Each team member's expertise contributed to a seamless integration of hardware and software, ensuring the system's precision and efficiency. The iterative process of testing, debugging, and optimizing reinforced the importance of resilience and innovation in overcoming technical challenges.

Beyond the technical accomplishments, the project emphasized the broader implications of agricultural automation, including reduced environmental impact, enhanced productivity, and cost efficiency. As a proof of concept, the tractor offers a scalable foundation for future development in autonomous farming technology. This collaboration with John Deere not only advanced our technical skills but also deepened our understanding of the transformative potential of robotics and digital systems in addressing global agricultural needs.

The project's success underscores the critical role of engineering in creating solutions that merge technological innovation with real-world applications, paving the way for advancements in smart farming and precision agriculture.

Reflection

Rosendo De Los Rios Moreno

Participating in this project allowed me to enhance my technical and teamwork skills significantly. My primary responsibility involved the mechanical assembly of the tractor and unit testing of hardware components. This role required meticulous attention to detail and an understanding of how different components interact. The challenges faced during assembly, such as ensuring stability and precise alignment of components, taught me the importance of problem-solving and adaptability.

This experience has inspired me to pursue further knowledge in embedded systems and mechanical integration, as I see their transformative potential in solving real-world problems. I am grateful for the opportunity to work on a project that bridges academic learning and practical application, and I look forward to contributing to similar initiatives in the future.

Jesus Garcia Gonzalez

As the lead for subsystem integration and testing, this project was an incredible learning opportunity that challenged and expanded my technical expertise. My focus on combining hardware and software components highlighted the complexities of ensuring seamless communication and functionality across the system. One of the biggest challenges was debugging integration issues, which required patience, critical thinking, and collaboration with my teammates.

I gained a deeper appreciation for the importance of rigorous testing and validation. Ensuring that all subsystems worked cohesively demanded not only technical skills but also a strategic approach to problem-solving. This process taught me that attention to detail and thorough documentation are critical in engineering projects.

Rodrigo Monterroso Bandy

My role in this project centered on motor control, debugging, and communication protocol implementation, all of which pushed me to refine my technical and problem-solving abilities. Working with PWM signals for motor control and integrating CAN protocols was both challenging and rewarding. These tasks required a strong understanding of hardware-software interactions and the ability to troubleshoot complex issues efficiently.

One of the most valuable lessons I learned was the importance of adaptability. While developing the motor control subsystem, unexpected complications arose that required creative solutions and collaboration with teammates. This experience reinforced my belief that innovation often stems from tackling challenges head-on.

Jordan Arturo Palafox Salinas

As the team member responsible for software development and communication framework implementation, this project was a significant learning experience. My focus on UART debugging and wireless communication protocols like SPI presented numerous technical challenges that required patience, precision, and collaboration to overcome.

The process of configuring the STM32 microcontroller and ensuring reliable data exchange across subsystems was particularly rewarding. These tasks demanded an in-depth understanding of embedded systems and pushed me to expand my skill set. One of the key takeaways was the importance of real-time responsiveness and the role of efficient software design in achieving system stability.

Acknowledgment

We extend our gratitude to our professors, Alfonso Avila Ortega, Alfredo Esquivel Jaramillo, Frumencio Olivas Alvarez, and Raúl Peña Ortega, for their guidance and support. Special thanks to John Deere for providing the opportunity to work on a project with real-world impact in advancing

smart farming technologies. This collaboration has enriched our learning and development, preparing us to tackle future engineering challenges.

Codes, documentation, and configurations

Link to drive where the final codes and configuration

 Codes, documentation, and configurations

Functional Tractor Overview

Link to drive where the tractor works in correct manners

 Video Tractor Embebidos